Shravya Manchala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.25
SBOA School and Junior College	Class XII (CBSE)	2021	95.60%
Paramita Heritage School	Class X (SSC - Telangana)	2019	98.00%

SKILLS

Tools: Cadence Virtuoso, Xilinx Vivado, LTSpice, MATLAB

Coding Languages: C, VHDL, Verilog

Hardware: Ardunio UNO, Arduino Nano, FPGA ZedBoard

RELEVANT COURSES

Digital Domain: Digital Electronics, Digital IC Design, Introduction to VLSI Design, Digital IC Design Lab

Analog Domain: Analog IC Design, Analog IC Design Lab

Devices Courses: Semi Conductor Device Modelling, Neuromorphic Devices for AI/ML

M. TECH THESIS

M. Tech Thesis Advisor: Prof. Shiv Govind Singh

Title: Electrochemical sensing of ovarian biomarkers using PCB based electrodes and read-out circuitry

- Electrochemical biosensing of ovarian biomarkers like apolipoprotein and HE4
- Development of PCB based electrochemical three-electrode system
- Development of read-out circuitry for Cyclic Voltammetry(CV) and Electrochemical Impedance Spectroscopy(EIS)
- Characterization of surface modification steps

PROJECTS

Implementation of 12 Hour Digital Clock Using Verilog

- Implemented the 12 hour digital clock with AM/PM indicator using the RTL code
- Hours ,Minutes and Seconds are incremented using the counters and the time is displayed in the format HH:MM:SS AM/PM

Real time data acquisition from ESP32 to FPGA

- Developed a real-time EMG data acquisition system using ESP32 and Zynq-7000 FPGA for sensor-data transfer and FPGA-side processing.
- Achieved 330 samples/s via UART and ~900 samples/s via Serial-to-Parallel transfer, with Vivado hardware configuration and Vitis software deployment.

RTL Design for Dual Clock Asynchronous FIFO

- Designed and implemented Dual Clock Asynchronous FIFO with different read and write clocks using RTL
- Employed Gray Encoding to avoid multi bit transistions and mitigate metastable issues

4-bit Binary-Weighted Current-Steering DAC

- Designed and simulated a 4-bit binary-weighted current-steering DAC using 8:4:2:1 weighted current branches for digital-to-analog conversion.
- Designed D-latch and latch-array circuits for synchronized switching and verified the complete DAC through schematic-level transient simulations.

PUBLICATIONS

IEEE: Compressor-Based Approach for Power-Optimized Error-Tolerant High Accurate Approximate Multipliers Design and Implementation

EXTRACURRICULAR

Hobbies: Playing Badminton, Drawing